Lesson Notes

MODULE 2 | LESSON 3

MEASURING THE PERFORMANCE OF STOCKS AND CRYPTOCURRENCIES

Reading Time	75 minutes
Prior Knowledge	Yield, Stocks, Cryptocurrencies
Keywords	Percent Return, Log Return, Holding Period Return, Price Range, Standard Deviation, Variance, Coefficient of Variation, Sharpe Ratio

In the previous lesson, we examined categories and comparisons of stocks and cryptocurrencies. In this lesson, we will measure how to compute the expected returns and standard deviations of returns. We will also spend time investigating whether we should use arithmetic or logarithmic returns.

1. Returns: An Introduction

The notion of return on an investment is fundamental for those who work in finance because a lot of what we do deals with returns. Returns are a measure of the performance of our investments and allow us to compare performances from different investments. Therefore, it is important to have a clear understanding of this notion and the related nuances.

1.1 Arithmetic Returns

Assume that we buy a share of company STV at time t_1 and sell it at time t_2 (obviously $t_2 > t_1$). What is the return from our investment in stock STV over the period $t_2 - t_1$?

If we call S_{t_1} and S_{t_2} the price of the stock at time t_1 and t_2, respectively, then the simplest formula to calculate the return from owning this stock over the period $t_2 - t_1$ is the following:

$$\textbf{arithmetic or percent return} := r_{t_1,t_2} = \frac{S_{t_2} - S_{t_1}}{S_{t_1}} = \frac{S_{t_2}}{S_{t_1}} - 1.$$

The formula we just wrote describes the *percent stock's price change* over a pre-specified time, which, in our setup, happens to be $t_2 - t_1$. This formula is not enough because stock S could pay a dividend over the period $t_2 - t_1$. To be specific, say that company S pays a dividend at time u where

$t_1 < u < t_2$. Then, if we call D_u the dividend paid at time u, the earlier formula should be changed to look like this:

$$\textbf{arithmetic or percent total return} := r_{t_1,t_2} = \frac{S_{t_2} + D_u - S_{t_1}}{S_{t_1}}.$$

This is not really a new result for us as we already saw it in an earlier lesson, but here, we want to stress as many nuances as possible in dealing with returns since we also want to provide a complete picture of how our investment in STV performed over the specified period of time $t_2 - t_1$.

Technically, we should also consider the fact that there are commissions and fees that should be included in the formula, which tend to be much smaller compared to capital gains and dividends, so we choose to ignore them at this point.

It is possible to compute total return as a dollar amount instead of a percentage. In this case, the formula is simply:

$$\textbf{dollar total return} = S_{t_2} + D_u - S_{t_1},$$

but we will mostly work with percent returns and percent total returns.

From now on, we will use the terms "total return" and "return" as equivalent, and the particular formula will tell us which is the case. In any case, when the stock pays no dividends, the two formulae are the same.

1.2 Logarithmic Returns

The second way to compute returns employs the following formula:

$$\textbf{logarithmic return} := \tilde{r}_{t_1,t_2} = \ln\left(\frac{S_{t_2}}{S_{t_1}}\right)$$

where $\ln(\cdot)$ is the natural logarithm function.

Note the use of the tilde symbol to differentiate the way the returns are computed in the two instances. The two formulae for percent return and log return tend to yield similar numbers, but they are not the same.

The formula can be easily extended to incorporate dividends, of course:

$$\textbf{logarithmic return} = \ln\left(\frac{S_{t_2} + D_u}{S_{t_1}}\right).$$

Example. *Suppose we bought a stock on January 2, 2024, paying $125. On February 28, 2024, the stock is valued at $153. Compute the arithmetic return and the logarithmic return.*

Answer. This is very simple:

$$\textbf{arithmetic return} = 153/125 - 1 = 22.40\%;$$

and

$$\text{logarithmic return} = \ln(153/125) = 20.21\%.$$

1.3 Logarithmic Returns or Arithmetic Returns?

The question of which of the two ways to compute returns is most appropriate comes up often. To answer this question, suppose that we invest an amount of S_0 dollars into an asset and we earn the following sequence of returns at time $t_1, t_2, \ldots, t_n$. In addition, we assume that $t_0 = 0$. Let's call these returns $r_1, r_2, \ldots, r_n$. Just to be sure that everyone is clear about the meaning, r_1 is the arithmetic or percent return between time 0 and time t_1, r_2 represents the percent return over the period between t_1 and t_2, and so on.

As investors, we are interested in the **cumulative return** between time 0 and time t_n. Then:

$$\text{cumulative return} := r_{0,t_n} = \frac{S_0 \times \prod_{i=1}^{n}(1 + r_i)}{S_0} - 1$$

and we observe that, if r_{0,t_n} represents the percent return between time 0 and t_n, then it must be:

$$r_{0,t_n} = \frac{S_{t_n}}{S_0} - 1 = \prod_{i=1}^{n}(1 + r_i) - 1.$$

Essentially, we are letting the returns compound continuously, i.e., we are reinvesting any gains between the time periods considered.

On the other hand, if we removed the profit in between periods so that at the beginning of each subperiod we always have an investment of S_0, then the cumulative return would be calculated as:

$$R_{0,t_n} = \frac{S_0 + S_0\left(\sum_{i=1}^{n} r_i\right)}{S_0} - 1 = \sum_{i=1}^{n} r_i.$$

Note the use of R instead of r to distinguish the two situations. R_{0,t_n} is the formula for total return without compounding and it is called the formula for total returns with **rebalancing**, meaning that if the return is positive, we take out the additional amount earned and spend it on something else (even use it to make another investment), and if the return is negative, we add money to our investment to bring it back to its original level of S_0. That is where the notion of re-balancing comes into play. Students have probably noticed the use of different symbols for total returns in the two instances as we are computing total returns differently.

You may wonder why you are being taken on this journey through cumulative returns. Be patient and you will soon find out. In any case, which type of cumulative return is the one that resembles the reality of investors most?

Let's rewrite the formula for the cumulative return with continuous compounding from above so that it is easier to discuss the implications:

$$r_{0,t_n} = \prod_{i=1}^{n} (1 + r_i) - 1,$$

or

$$1 + r_{0,t_n} = \prod_{i=1}^{n} (1 + r_i).$$

Let's take the natural logarithm on both sides of the equation, which, using simple properties of logarithms, yields

$$\ln(1 + r_{0,t_n}) = \sum_{i=1}^{n} \ln(1 + r_i).$$

Finally, recall that $r_i = S_i/S_{i-1} - 1$, and if we replace this formula in the one immediately above, we end up with

$$\ln(1 + r_{0,t_n}) = \sum_{i=1}^{n} \ln\left(\frac{S_i}{S_{i-1}}\right).$$

Furthermore, we have also

$$\ln(1 + r_{0,t_n}) = \ln\left(1 + \frac{S_{t_n}}{S_0} - 1\right) = \ln\left(\frac{S_{t_n}}{S_0}\right) = \tilde{r}_{0,t_n}.$$

In other words, we have shown that the total return using continuous compounding is equal to the sum of the log returns in the subperiods, i.e.:

$$\tilde{r}_{0,t_n} = \sum_{i=1}^{n} \tilde{r}_{t_{i-1},t_i}.$$

and this property is one of the reasons for preferring the log returns. However, since

$$\ln(1 + x) \approx x$$

when x is close to zero, when the returns are sufficiently small, the two ways to compute the returns tend to produce similar results.

Another argument for favoring log returns over percent returns is that if prices are log-normally distributed, then returns would be normally distributed. But this is something important for several models used in quantitative finance that will be explained in detail in the Derivative Pricing course.

2. Return: Time-Specific Returns

Suppose we are given daily closing prices, i.e., the time indices t represent days. We can easily calculate daily returns, and we already know how to do it. However, say that we wanted to compute weekly returns from these daily data points. We can do this several ways; for instance, we could calculate returns using the closing prices on Friday's:

$$\text{weekly return week t+1} = \ln\left(\frac{\text{Fridays price week t+1}}{\text{Fridays price week t}}\right).$$

There is nothing special about Friday's and we could have done the same using Mondays, Tuesdays, etc. Likewise, you can do the same for monthly returns:

$$\text{monthly return month t+1} = \ln\left(\frac{\text{price last trading day in month t+1}}{\text{price last trading day in month t}}\right).$$

Months have different numbers of days, but it is typical to use the last trading day of the month for these calculations. We can calculate quarterly and annual returns as well and, in principle, returns for any timeframe we like. For quarterly returns, we could use March 31, June 30, September 30, and December 31. Likewise, annual returns typically use the values from December 31 in one year to another.

Note that if we had monthly price data, we could not compute weekly or daily returns.

It is worth saying that most closing prices for stocks are only collected from Monday to Friday when the stocks are actively traded. On Saturdays and Sundays or on holidays, because the markets are closed, there is no data, but data providers can simply extend the data by setting Saturday and Sunday prices equal to the price on Friday. This could create some problems when working with daily data because this choice may reduce the volatility in our dataset arbitrarily. In contrast, it is less of a problem when working with weekly or monthly data. Assets like cryptocurrencies do not have this problem as they can trade 24/7. When using both closing prices for stocks and cryptocurrencies, for example, we must pay attention to the different availability of data points.

An important observation has to do with the fact that if we start with N daily data for closing prices and we want to compute daily returns, then we would be able to compute only $N-1$ daily returns. We are not able to compute the return for the first day, as we would need an extra price before the first day in our list. Also, note that if we wanted to compute weekly returns from daily closing prices and we start with N closing prices, we would be able to compute about $N/7$ returns. To obtain reliable estimates for the parameters we need in many of the models we use in finance, we require a good number of data points, so we have to consider these aspects. It should be evident that if we go for yearly returns, we need data over a very long time. While getting a long series of yearly closing prices is not a problem, we have to consider that if we use the data for modeling, the underlying models could have changed over time.

We could compute returns over different time intervals, but it would be quite complicated to compare them. Imagine that one trader takes pride in his 3% monthly returns and that another trader parades her 20% yearly returns. Who is doing better? It is typical for funds to send their investors reports about the performance of the investors' assets with the fund. Reports can be sent quarterly, semi-annually, or annually. If we received a quarterly report saying that in the most recent quarter our account had a return of 5.5%, we would probably want to know how much that is in terms of an annual return. The need to make all returns directly comparable often drives us to annualize returns.

Example. *Suppose your assets in Fund X were $100,000 on July 1. On December 31, your assets are worth $110,000. What is the return? Well, there's a 10% gain on your $100,000, but it only took 6 months. To annualize the return, you would effectively multiply by a factor that uses a 12-month hold. Since there are 12/6 = 2 (two) 6-month periods in 12 months, we simply double the 6-month return. Therefore, it is a 20% gain annualized.*

The main criticism of annualized returns that are obtained via an extrapolation, like the one in our example above, is that they may not reflect actual P&L.

Example. *We got lucky and bought a stock yesterday for $2 and are able to sell it today for $2.20. That is a return of 10% in one day!! We think we are finance gurus and go around telling everyone (who cares to listen to us) that we achieved an annualized return of 3650%. "Warren Buffett who?" However, we should recognize that making 10% in one day does not indicate that we have otherworldly abilities, and we cannot be expected to repeat the same return hundreds of times. Perhaps tomorrow, the price of our stock goes down to $1.95, and we would have to advertise a negative return instead.*

The last example makes it clear that we have to distinguish an actual annual return, which is based on points that are truly one year apart, from a return that has been annualized. As a matter of fact, certain report standards, like GIPS (Global Investment Performance Standards), forbid the use of annualized returns, or the report must clearly state that they are annualized and not actual returns. In fact, when evaluating the performance of funds, we may have to pay attention to another "trick" that is sometimes employed: reporting only enough years to show the best returns and excluding years where the fund did not do too well.

Example. *Suppose that a firm that started on November 1, 2023, reports a 2-month initial return of 4% on December 31, 2023, which equates to an annualized return of 26.53% (using compound interest, the most suitable for this purpose). According to GIPS (Global Investment Performance Standards) standards, the firm is required to present the 4% as a partial year's performance. The annualized return of 26.53% must not be presented. Suppose that the firm does not follow GIPS to report its performance and presents the annualized return. What should a savvy investor do?*

Answer. The best approach for a savvy investor who is aware of the difference between actual performance and on-paper performance would be to de-annualize the return that was presented. Therefore, the investor can compute the actual 2-month performance as:

$$(1 + 26.53\%)^{2/12} - 1) = 0.04 \text{ or } 4.00\%.$$

Let's consider another return related to the actual time the investment is held: the **holding period return**, which is often used by investors who simply like to understand how much they made on an investment. Suppose we put $100,000 in an investment and then sell it at $170,000. We made a 70% return, yet it's unclear how long it took to earn it: anywhere from seven days to many years. Holding return is not annualized; it merely represents whether a profit was made.

One final return to consider is the **net return**. This will be inclusive of all incomes and costs. Suppose we've bought a stock at $100 plus a $2 commission, received a $7 dividend, and then sold it for $115. What is the net return? The formula to use can be described as

$$\textbf{Net Return} = \frac{\textbf{(money earned from investment–money invested)}}{\textbf{money invested}}$$

which for our investment becomes

$$\textbf{Net Return} = \frac{\mathbf{(115+7)-(100+\$2)}}{\mathbf{(100+2)}} = \mathbf{20/102 = 19.6\%}$$

Pay attention to the fact that the $2 commission is the cost to enter this investment and must be included among the outflows of cash.

In addition, notice that we have not discussed taxes here. Certainly, taxes will eat into net returns. Because this class is intended for a global audience, tax rates vary wildly from country to country and even within a country from one bracket to another. As such, no discussion of taxes will be included here, but keep in mind that taxes will only lower the actual returns from these investments.

3. Moving from Returns to Volatility

We have focused on what we are expected to make, so we can use return and its calculations to estimate what our gain is going to be. Surely if our expected returns were negative, meaning we would expect to lose money, we would likely not want to invest much, if anything, in such a venture. Buying a lottery ticket for $2 is an exception because the loss is very small, but the potential win, while quite improbable, would be incredible! But in general, we would not expect to invest in that security.

Since we are investing in some type of risky asset, we're not sure if we're going to achieve that return. We can use historical returns to estimate what our future return is going to be. However, this expected return is nothing more than an average. When you think about an average, you're really thinking of an expected value from a random distribution. Since the distribution of results can vary, it would help to know what the expected distance is we are likely to be from that average.

In other words, even if we don't get the expected value, are we likely to be close? We are interested in the volatility around the average. This is known as the **standard deviation**.

4. Volatility: An Introduction

Standard deviation becomes the central measure of volatility but is just one way to measure volatility. The concept of volatility is much broader. It essentially addresses the fluctuation of the series itself. There are entire books written as to why volatility exists and how it can be modeled and forecasted in financial markets. Keep in mind that markets are complex systems because there are many different participants that are operating in different economies and using different models, with different tolerances for risk, cash flow needs, tax consequences (e.g., long-term capital gains are taxed differently than short-term capital gains), etc.

Therefore, the emergent property of volatility can often defy simplistic modeling that distills volatility down to a simple standard deviation. In this program, we will continually develop and refine our understanding of volatility. For now, let's consider that at any given time, there are market participants who feel a security is ready to be sold and others who feel it is the right time to purchase--that is, what helps to facilitate trading. Consequently, market prices fluctuate.

A big influence on market volatility is news. Impactful news does not arrive equally throughout the day, will vary in its importance, and will even influence some participants to make very different decisions from others. Some of these participants may be algorithms that respond to the sentiment of headlines and frantically sell shares as a result. Others may be professional managers who make the same decisions but more slowly. Finally, there are others that pause and reflect to determine if those same securities are worth purchasing at such values. One reason that volatility can change is that financial news is not absorbed in the market equally by different participants making varied decisions on different time scales.

5. Volatility: Measures

Indeed, what appears as noise to some participants can be trading opportunities for others. Consider prices that have "jumpiness" between two levels. Throughout the trading day, there is a high price at which trading occurs, "the high," and a low price, "the low." One way to measure volatility is to simply measure this price range. Price range is our first measure of volatility. It is in the units of price.

$$\textbf{Daily Price Range} = \textbf{Daily High Price–Daily Low Price}$$

Example. *Suppose Friday's stock price reached a high of $105 and a low of $99. Then Friday's price range is $6. Suppose further that Monday's stock price reached a high of $106 and a low of $104; Monday's price range is $2. We would agree that, using price range, Friday was more volatile than Monday was. Using price range, the volatility is measured in units of price, which is how the stock trades.*

To calculate the price range, we need to know the day's high and low prices. Merely knowing closing prices would be insufficient to determine the range. Throughout the trading day, think of the high as the highest point reached--the high-water mark--and the low as the lowest point reached--the low-water mark. Price range volatility looks at the difference between those two extremes--the two water marks. A very lucky day trader would have bought at the day's low price (low-water mark) and sold at the day's high price (high-water mark), so the intraday range is the maximum amount the trader could have made day trading the stock.

Another way to look at volatility is to use returns instead of prices. If we had daily returns, then we could look at the last 20 days of returns and compute their standard deviation. That means over 21 trading days (remember you need 21 days to get 20 returns), which in the New York Stock Exchange equates to about one month of trading, you would basically be calculating the standard deviation of the daily returns for one month. Stocks that are very volatile will have a high standard deviation of return. Stocks that have a low standard deviation will not show as much fluctuation.

When we use returns to compute standard deviation, the units of that standard deviation will be in return units rather than price. This gives a different perspective of volatility than price range does. Standard deviation is the expected distance you are from the average.

Suppose the average daily returns were 0.2% and the standard deviation was 0.5%. If we agree or just assume that returns are normally distributed, then we can say that 68% of the returns are within 0.2% - 0.5% on the low end to 0.2% + 0.5% on the high end. This equates to the return being between -0.3% and 0.7% for 68% of the time.

Variance is the square of standard deviation. Sometimes, people use variance and standard deviation interchangeably. Please avoid doing so.

Variance is another way to think of volatility, but its units are squared returns. In other words, variance is the average squared deviation from the average.

The problem stems from the fact that deviations can be positive or negative. A return that is below the average is a negative deviation. A return that is larger than the average is a positive deviation. When we want to understand fluctuation or volatility, we are focused on the size of the deviation and not the sign. Therefore, we need to make the negative and positive signs disappear. The easy way of doing that is to square them. (Taking absolute values would be another way). This is exactly what variance measures: squared deviations. To get back to the correct units, distance, we need to take the square root. Stated in the other direction, standard deviation is the square root of variance. It gives the expected distance we will be from the expected value. When people discuss volatility, they are likely referring to standard deviations rather than variances, since they have the same units of return. If you had a 20% standard deviation when you square 20%, you get a 4% variance. Just be sure that when you use either one, you're conscious of which one you're utilizing.

6. Volatility Divided by Return: The Coefficient of Variation

We have discussed returns. We have discussed volatility. How can we use these to make investment decisions? Ideally, we would like to have a lot of return for as little volatility as possible. More expected return means higher profits. Less volatility means we are more likely to achieve those high profits. Lowering returns reduces our profits. Raising volatility means we increase the likelihood of being far below those profits to the point where they become losses!

So let's combine the two. First, we'll start by dividing one by the other. We'll start by dividing volatility by return. Volatility divided by return is known as the coefficient of variation. Volatility is the numerator and return is the denominator. This is a classical metric from statistics. In finance, when we compute the coefficient of variation, we are basically asking the following question: how much volatility does an investment encounter to earn a given amount of return? That is, you are effectively seeing how much volatility costs per unit of return. Think of it as an exchange rate. You have to spend x amount of volatility dollars to earn y amount of return units.

Suppose you use this ratio to compare the relative risk/return relationship of two investments, such as two stocks. Stock A has a 50% volatility per unit of return. Stock B has 25% volatility per unit of return. Which of these seems like a better investment? We've intentionally not provided the specific returns and volatilities, just this ratio. This is important because the returns of the two stocks can be very different, as well as their volatility. This ratio can be thought of as using a volatility currency to purchase a return currency. How much volatility do you have to spend to earn a given amount of return? For Stock A, you have to spend 50 volatility units. For Stock B, you only have to spend 25 volatility units. Relatively speaking, Stock B is a bargain. You do not have to spend, that is, encounter, as much volatility to earn the same amount of return. The lower the coefficient of variation, the more preferable the stock is because you don't expect to face as much volatility to earn the same amount of return that you expect for the other.

Note that this assumes you had good samples of the data, that these are representative of the future price changes, and that you did your calculations correctly. All these assumptions can be questioned and perhaps even validated.

7. Return Divided by Volatility: Sharpe Ratio

Suppose instead of dividing volatility by return, we change the order and divide return by volatility. This is the more classical way to approach the problem. In this case, now, we have return as the numerator and volatility as the denominator. Suppose we change the numerator to reflect the market premium:

$$\textbf{Market Premium} = \textbf{Stock Return–Risk-Free Rate}$$

The idea combines not only return and volatility but also the fixed-income market. The fixed-income market is important because it shows what any risk-adverse investor would earn by simply being in the most conservative investment: the risk-free rate. The risk-free rate is the rate earned simply for investing in a risk-free bond. Since stocks are risky, they should earn returns in excess of that return. If a stock were to earn less than the risk-free return, then a rational investor would forgo the volatility of the stock market and stick with risk-free investments, which protect the principal while still earning some return.

The numerator in our example is the market premium. So, suppose for stock C, the market premium and standard deviation are 7% and 10%, respectively. Suppose for stock D, the market premium and standard deviation are 8% and 12%, respectively. Which is the preferred investment?

$$\textbf{Stock D:} = \frac{\textbf{Market Premium}}{\textbf{Volatility}} = \mathbf{8/12 = 0.67 \ \text{or} \ 67\%.}$$

In this case, stock C has a higher return per unit of volatility (risk) than stock D does. We would prefer the investment that has a higher amount of return per unit of risk. Now, this is the opposite of our coefficient of variation. When we express volatility units per return, we would like the lower number. When we express return units per volatility, we prefer the larger number. Here, the larger number means we expect to earn more return per unit of volatility.

In this course, we'll name this ratio (keep a *sharp* lookout for it) and use it extensively to compare investments, but for now, the important thing is that we've seen how return and volatility can be combined to compare investments. This can just as easily work if these securities are from different asset classes. Return and risk help encapsulate the key statistical properties of a financial asset's performance. In the next lesson, we'll dive deeper into the details of these statistics.

8. Conclusion

In this lesson, we measured returns and standard deviations. These measures tell us what we expect to earn (or lose) and how volatile the actual value can be. In the next lesson, we'll try to assess if these returns come from a distribution that resembles a Gaussian or normal distribution.